

SATYAM SAKRAL

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PROFESSIONAL SUMMARY

Software and AI/ML Engineer (MCA Candidate, Class of 2026) with hands-on experience building backend REST APIs, real-time web applications, and applied machine learning models. Skilled in Python, Java, SQL, Django, FastAPI, PyTorch, and LangChain RAG systems. Experienced in query optimization, data pipeline engineering, and LLM fine-tuning to deliver efficient, production-ready software solutions.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

AI / ML & LLMs: Retrieval-Augmented Generation (RAG), Fine-Tuning, Prompt Engineering, LangChain, PyTorch, OpenCV, Scikit-learn, Model Evaluation

Web & Backend: FastAPI, Django, Spring Boot, RESTful APIs, WebRTC, Socket.io

Databases: PostgreSQL, Supabase (RLS), MySQL, MongoDB

Tools & Workflows: Git, GitHub, Docker, Postman, Semantic Search, Vector Embeddings

WORK EXPERIENCE

Ethara AI — LLM Trainer Intern

Apr 2024 – Jun 2024

- **LLM Post-Training & Optimization:** Assisted in fine-tuning, RLHF alignment, and evaluation workflows for large language models to improve response accuracy and task coherence.
- **Dataset Engineering & Quality Control:** Built data processing pipelines to clean, filter, and deduplicate multi-modal datasets, reducing dataset noise and model hallucination rates.
- **Flux AI Framework Integration:** Contributed to evaluation benchmarks and feedback loops within core Flux AI framework pipelines.

Doosra College — Django Developer Intern

Jul 2023 – Sep 2023

- **Full-Stack Inventory Backend:** Developed a centralized Inventory Management System using Django REST framework and SQL to streamline logistics tracking across departmental modules.
- **Database Optimization:** Re-engineered complex SQL queries and database indexing, achieving a **60% reduction** in data retrieval latency for high-throughput operations.
- **Analytics Dashboard Integration:** Worked with cross-functional teams to integrate interactive data visualization dashboards into the operational pipeline.

PROJECTS

Study AI – Adaptive Learning & Neural RAG Platform | [GitHub](#)

Python | FastAPI | LangChain | Supabase

- **Zero-Trust BYOK Architecture:** Built an AI Gateway supporting OpenAI, Anthropic, Gemini, Groq, and OpenRouter using Fernet AES-GCM encryption for secure client API key storage.
- **Multi-Source Neural RAG Pipeline:** Implemented multi-format document parsing (PDF, DOCX, PPTX, YouTube Transcripts) with vector semantic search for context-aware document Q&A.
- **Adaptive Planner & Dynamic Assessments:** Developed study plan scheduling algorithms based on topic difficulty and learning speed, paired with automated exam feedback generation.

FaceMask Detection – Real-Time Computer Vision Classifier | [GitHub](#)

Python | PyTorch | OpenCV

- **CNN Architecture & Inference Pipeline:** Built and trained a Convolutional Neural Network (CNN) for real-time facial mask classification on live video feeds (~81% accuracy).
- **Data Processing & Augmentation:** Engineered image pre-processing and data augmentation pipelines (scaling, rotation, lighting adjustments) to maintain inference stability under varied environment conditions.

VividChat – WebRTC Real-Time Video & Messaging System | [GitHub](#)

Django | WebRTC | Socket.io

- **Peer-to-Peer Streaming Architecture:** Developed a low-latency video streaming web app using Django backend controllers and WebRTC peer connection protocol.
- **Scalable Signaling Infrastructure:** Integrated Socket.io for dynamic room management, group calling capabilities, and real-time text broadcasting.

EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU)

Delhi, India

- **Master of Computer Applications (MCA)** – JIMS Rohini | CGPA: **8.2 / 10** | *Expected: Jun 2026*
- **Bachelor of Computer Applications (BCA)** – SGTBIMIT | CGPA: **7.7 / 10** | *Graduated: Jun 2024*

CERTIFICATIONS

- **Specialized Program in AI & ML with Drone Tech** – TiHAN, IIT Hyderabad
- **Java Programming Masterclass** – Udemy Certification